Replicating Antman and Cortes (2023): The Long-Run Impacts of Mexican American School Desegregation

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Abstract

Antman and Cortes (2023) examine long-run educational attainment for Hispanics and non-Hispanic Whites in California after the 1947 *Mendez v. Westminster* court decision ended de jure segregation of Mexican American schoolchildren. They use a difference-in-differences design across high and low segregation counties. In this document, I replicate Figure 1, Table 1, Table 2 Panels A–C, and Figure 2. I extend Figure 2 with event studies for the remaining two outcomes. Sample sizes are exact and point estimates differ only slightly, consistent with IPUMS revisions.

1 Data and Empirical Strategy

(Antman and Cortes, 2023, p. 892) use IPUMS 5 percent public use samples of the 1980, 1990, and 2000 US decennial censuses (Ruggles et al., 2024). The sample consists only of people born in California. Years of schooling is built from IPUMS EDUCD. All estimates are unweighted, discussed in Section 3.3.

Likely segregation is measured using Hispanic to non-Hispanic population ratios for all California counties in 1940. The difference-in-differences design compares HiSeg (counties above the seventy-fifth percentile) with LoSeg (counties below the twenty-fifth percentile) (p. 893). The treatment group is ages 2–6 (born 1941–45, $PostMendez_t = 1$), and the comparison group is ages 12–16 (born 1931–35, $PostMendez_t = 0$) in 1947. The 1936–1940 group is dropped because it would have been partially treated, similar to Duflo (2001) (p. 897). $HiSeg_c$ is the same for everyone in a county, so standard errors are clustered at the county level.

$$Y_{ict} = \alpha + \pi \left(HiSeg_c \cdot PostMendez_t\right) + X_{ict}\,\beta + \mu_c + \delta_t + \varepsilon_{ict} \tag{1}$$

Equation (1) is an OLS regression with two-way fixed effects (p. 896). π is the difference-in-differences estimate. Y_{ict} is the educational outcome of person i in county c, born in year t. There are three outcomes: years of education, completing junior high, and completing high school. X is a vector of controls. μ_c are county fixed effects and δ_t are birth-year fixed effects. ε_{ict} is the error term.

*Code, data instructions, and outputs: `https://github.com/iellahi/Antman_Cortes_Repl`.

2 Results[1]

2.1 Educational attainment by cohort

Figure 1 shows average years of education by birth cohort, plotted by age in 1947. The rise for HiSeg Hispanics is clear. It begins with cohorts aged ∼17 in 1947 and continues through cohorts still in school. This is consistent with desegregation raising schooling. The county gap for Whites is much smaller than Hispanics. The series converges for the youngest cohorts, the two White series meet and the Hispanic gap is at its smallest. The figure uses the same series, line styles, axis and legend as the published figure (p. 894).

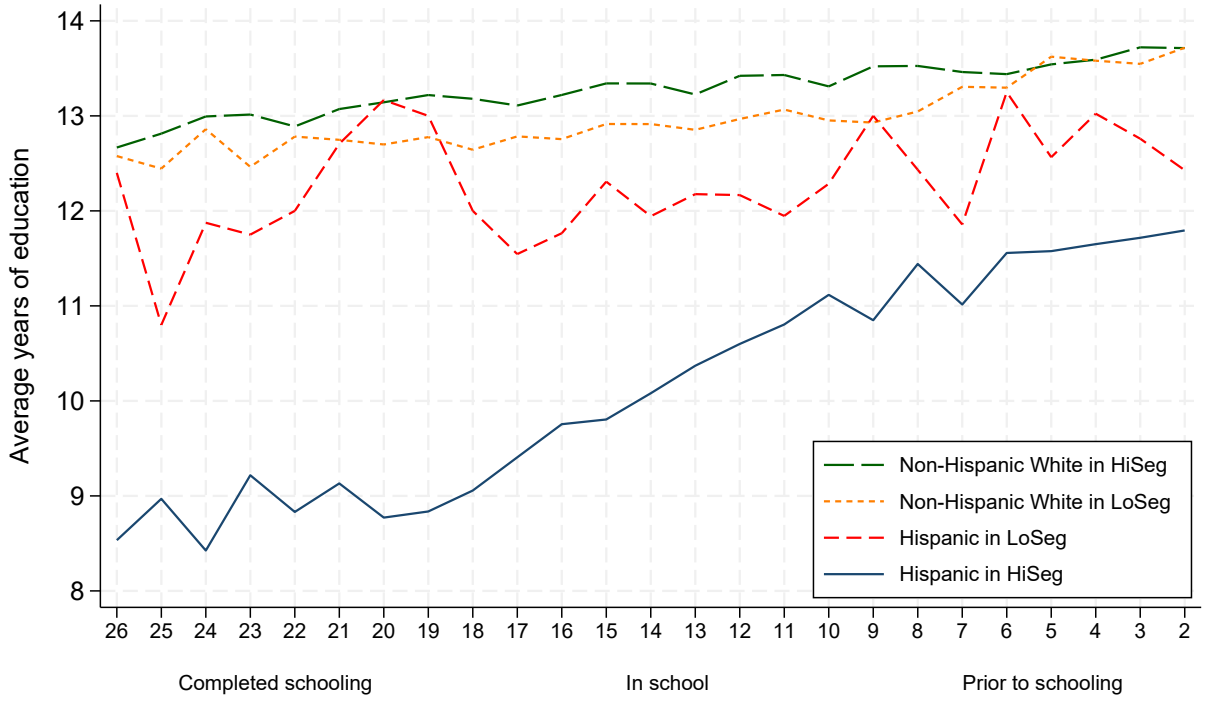


Figure 1: Educational Attainment in High- and Low-Segregation Counties by Age at *Mendez v. Westminster* School Desegregation Decision

2.2 Summary statistics and the raw difference-in-differences

Table 1 shows mean outcomes by county group and cohort, the HiSeg–LoSeg gaps, and the raw difference-in-differences. All sample sizes and all junior high and high school values are identical to the published table (p. 896). There are small differences in years of education means, SDs, and the DiD in the second and third decimal. This is likely due to IPUMS revisions to education codes above 12 years. Binary outcomes cut at 8 and 12 years are unaffected.

[1]Figure and table titles in this section are reproduced verbatim from Antman and Cortes (2023).

Table 1: Sample Descriptive Statistics on Mexican American Educational Attainment by Intent-to-Treat Status

	Panel A. Treatment group 1941–45 birth cohorts (ages 2–6)			Panel B. Comparison group 1931–35 birth cohorts (ages 12–16)			
	High	Low	$\Delta^{\text{treatment}}$	High	Low	$\Delta^{\text{comparison}}$	$\Delta^{\text{treatment}} - \Delta^{\text{comparison}}$
Years of Education	11.669	12.825	-1.156***	10.129	12.052	-1.923***	0.767**
	[2.891]	[2.587]	(0.222)	[3.856]	[2.311]	(0.274)	(0.353)
Jr. High (8+ years of education)	0.929	0.979	-0.051***	0.777	0.974	-0.197***	0.146***
	[0.258]	[0.144]	(0.013)	[0.416]	[0.160]	(0.020)	(0.024)
High School (12+ years of education)	0.669	0.811	-0.142***	0.479	0.701	-0.223***	0.080
	[0.471]	[0.393]	(0.034)	[0.500]	[0.461]	(0.053)	(0.063)
Number of observations	3,217	143	3,360	2,193	77	2,270	5,630

2.3 Regression estimates

Table 2a shows the main estimates of equation (1). Table 2b shows the placebo test on cohorts born 1921–30 (too old to benefit); no estimate is positive. Table 2c shows small negative estimates for non-Hispanic Whites. Minor differences from published estimates (p. 899, within one standard error) are likely driven by the years-of-education levels as well.

2.4 Event study

In Equation (2) below, a set of birth-year indicators replaces $PostMendez_t$. π_τ is the difference between HiSeg and LoSeg for cohort τ. Figure 2 shows negative estimates for cohorts aged 18–26 in 1947, estimates near zero for ages 12–17, and positive estimates for ages 2–11 (p. 900). The reference cohort is age 16 in 1947, marked by †. Confidence intervals are wide because each cohort's LoSeg sample is small.

$$Y_{ict} = \alpha + \sum_{\substack{\tau=1921 \\ \tau \neq 1931}}^{1945} \pi_\tau \left(HiSeg_c \cdot \mathbf{1}[t = \tau] \right) + X_{ict}\,\beta + \mu_c + \delta_t + \varepsilon_{ict} \tag{2}$$

Table 2: The Impact of *Mendez v. Westminster* School Desegregation Decision on Educational Attainment

(a) Panel A. Hispanic Sample

	(1) Years of education	(2) Jr. High School	(3) High School
High segregation × post-Mendez cohort	0.913**	0.153***	0.090*
	(0.352)	(0.035)	(0.044)
Mean (y-variable)	11.104	0.872	0.599
Number of observations	5,630	5,630	5,630
R^2	0.107	0.103	0.069
Birth cohort fixed effects	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes

(b) Panel B. Placebo Sample (Hispanic Birth Cohorts 1921–1930)

	(1) Years of education	(2) Jr. High School	(3) High School
High segregation × placebo post-Mendez cohort	-0.414	-0.007	-0.133*
	(0.421)	(0.063)	(0.069)
Mean (y-variable)	9.042	0.681	0.350
Number of observations	3,498	3,498	3,498
R^2	0.087	0.091	0.045
Birth cohort fixed effects	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes

(c) Panel C. Non-Hispanic White Sample

	(1) Years of education	(2) Jr. High School	(3) High School
High segregation × post-Mendez cohort	-0.391***	-0.009**	-0.026**
	(0.080)	(0.004)	(0.010)
Mean (y-variable)	13.474	0.986	0.895
Number of observations	26,884	26,884	26,884
R^2	0.056	0.008	0.019
Birth cohort fixed effects	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes

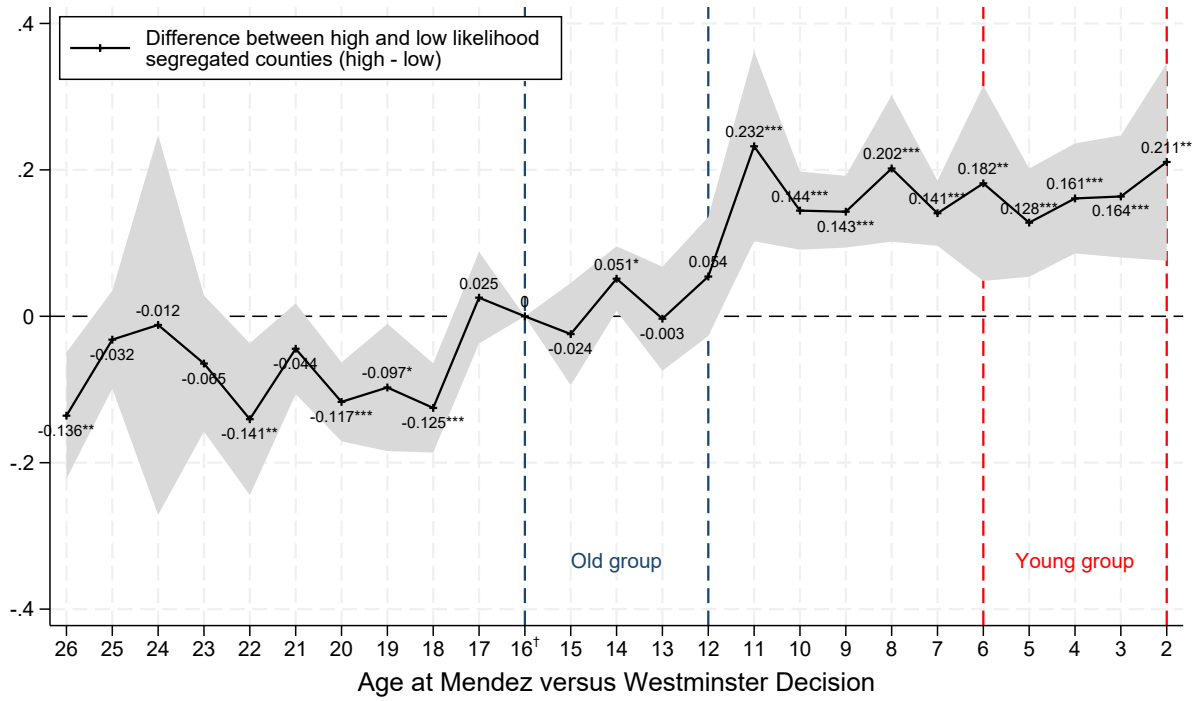


Figure 2: Event Study Analysis—Educational Attainment for Hispanics, Junior High School Outcome

3 Discussion

3.1 Threats to identification

Parallel trends. The design relies on the assumption that without the *Mendez* decision, HiSeg and LoSeg areas would have moved together identically. A shock that threatens both cohorts differently is possible, even as fixed effects absorb county and cohort level shocks (e.g., farm labor demand under the Bracero program (1942–64) concentrated in HiSeg Central Valley counties). Existing tests do support parallel trends. Table 2b (placebo) has no positive estimates. In Figure 2 (event study), ages 12–17 are near zero and show no drift before the treated cohorts. At the same time, however, these tests do not address migration.

Endogenous migration. A county is the place people lived in 1980–2000 at ages 35–79. Families who moved after 1947 can affect the treatment variable. The direction of this selective sorting effect is ambiguous. It could be the case that educated Hispanics moved into HiSeg areas like Orange County to take advantage of the policy, pushing up the estimate. At the same time, educated Hispanics could have moved out of rural HiSeg areas, pushing it down. Childhood county-level data from the 1950 census linked to adult outcomes is one piece of data that could help. Ethnic attrition is another possibility, with some Hispanics no longer identifying as Hispanic (Duncan and Trejo, 2011).

3.2 A continuous-treatment alternative

An alternative could involve interacting a continuous 1940 Hispanic share (instead of the HiSeg/ LoSeg) with $PostMendez_t$ using every identified county.

Advantages. The increased sample size allows for increased statistical power. There are no arbitrary percentile cutoffs. Estimates are then generalized across the entire distribution of counties, not just the extremes. Counties like Los Angeles and San Diego come back into the fold.

Disadvantages. Such a model assumes linearity: a 1% increase in Hispanic population share has a constant effect on segregation likelihood regardless of the baseline. Segregation likely follows a tipping point model. A certain critical mass of Hispanics is needed for a school district to invest in separate infrastructure. Middle counties excluded by the authors likely had fuzzy segregation practices, which means that they can add measurement error. Huge counties like Los Angeles and San Diego would dominate the estimate, though the paper does report that adding them to HiSeg changes little (p. 894, fn. 16). The authors' quartile approach likely maximizes the contrast between treated and untreated areas.

3.3 Sampling weights

The 1980 sample is self-weighting, every person a weight of 20. The 1990 and 2000 samples have varying weights (0–117 and 0–129) because long-form sampling rates differed by place size and responses were adjusted to population totals.

Point estimates. Unweighted OLS estimates the average effect on a sampled person. If effects differ for different people, weighted and unweighted estimates may not represent the true average treatment effect for Hispanics (Solon et al., 2015). If the impact of desegregation was heterogeneous (e.g., different in rural areas than in urban ones), estimates would over or understate the aggregate benefit.

Standard errors. Avoiding weights can maximize efficiency. Under the Gauss–Markov theorem, if the error terms are homoskedastic, unweighted OLS provides the best linear unbiased estimator (BLUE) with the lowest variance. Clustered standard errors already address heteroskedasticity. The small number of clusters presents a Type I error risk.

4 Conclusion

The replication succeeds: sample sizes are exact and estimates are within IPUMS revision noise. The event-study extension adds the same pattern for years of education and a weaker one for high school (Figures A1 and A2). Section 3.1 points to limitations.

A Additional event-study results

Figure A1 shows the event study for years of education. Figure A2 shows it for high school completion. Table A1 lists every coefficient behind Figures 2, A1, and A2.

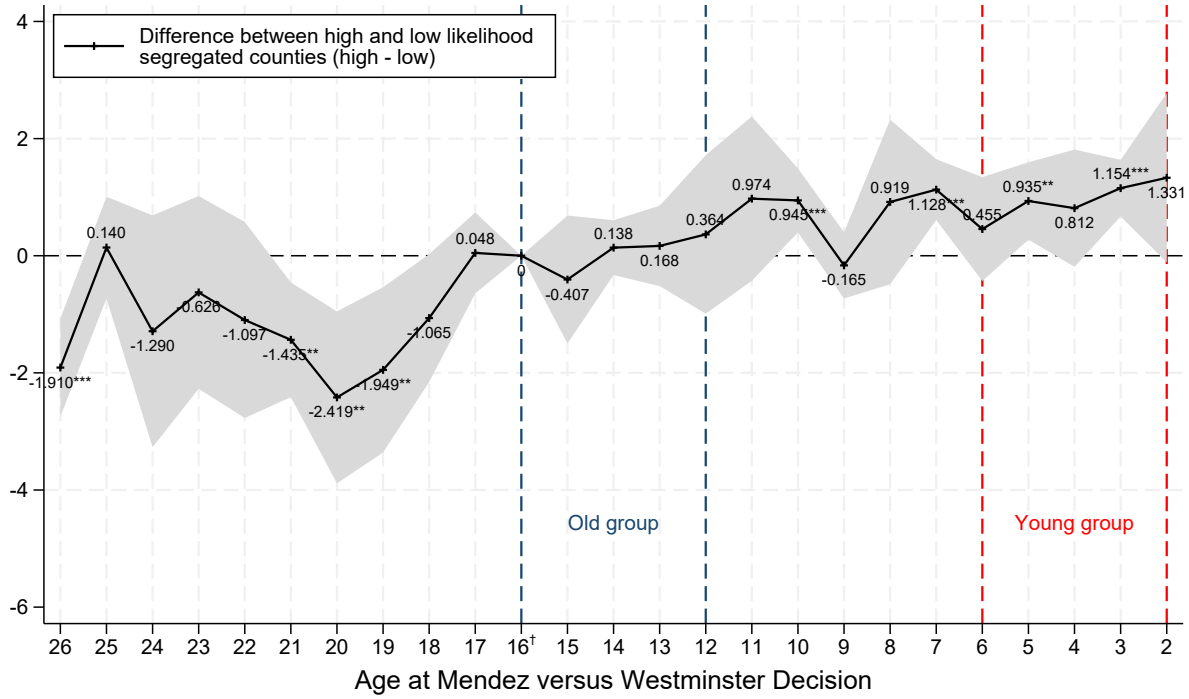


Figure A1: Event Study Analysis—Educational Attainment for Hispanics, Years of Education Outcome

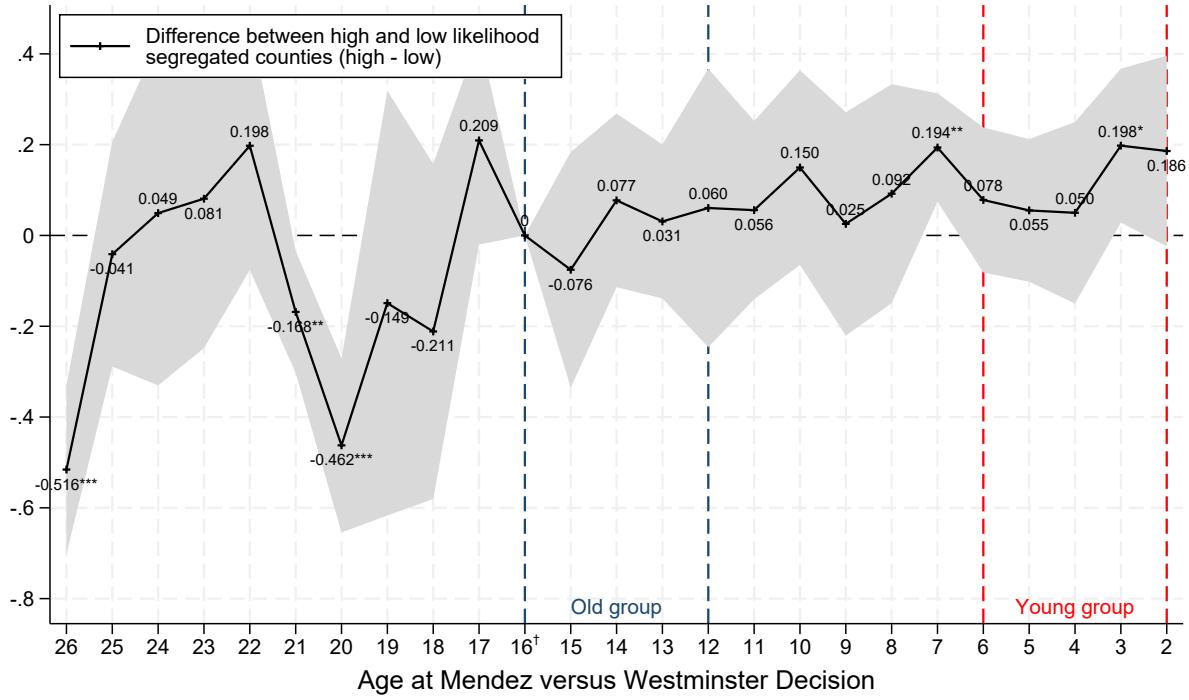


Figure A2: Event Study Analysis—Educational Attainment for Hispanics, High School Outcome

Table A1: Event Study Coefficients by Birth Cohort, All Three Outcomes

		Jr. high (8+)		Years of education		High school (12+)	
Age in 1947	Birth year	Coef.	SE	Coef.	SE	Coef.	SE
26	1921	-0.136**	(0.050)	-1.910***	(0.481)	-0.516***	(0.107)
25	1922	-0.032	(0.038)	0.140	(0.499)	-0.041	(0.142)
24	1923	-0.012	(0.149)	-1.290	(1.137)	0.049	(0.218)
23	1924	-0.065	(0.053)	-0.626	(0.946)	0.081	(0.189)
22	1925	-0.141**	(0.060)	-1.097	(0.962)	0.198	(0.157)
21	1926	-0.044	(0.036)	-1.435**	(0.563)	-0.168**	(0.077)
20	1927	-0.117***	(0.031)	-2.419**	(0.843)	-0.462***	(0.110)
19	1928	-0.097*	(0.050)	-1.949**	(0.811)	-0.149	(0.269)
18	1929	-0.125***	(0.035)	-1.065	(0.627)	-0.211	(0.212)
17	1930	0.025	(0.036)	0.048	(0.397)	0.209	(0.132)
16	1931	0		0		0	
15	1932	-0.024	(0.040)	-0.407	(0.628)	-0.076	(0.149)
14	1933	0.051*	(0.025)	0.138	(0.269)	0.077	(0.110)
13	1934	-0.003	(0.041)	0.168	(0.394)	0.031	(0.097)
12	1935	0.054	(0.047)	0.364	(0.778)	0.060	(0.176)
11	1936	0.232***	(0.075)	0.974	(0.807)	0.056	(0.113)
10	1937	0.144***	(0.031)	0.945***	(0.314)	0.150	(0.123)
9	1938	0.143***	(0.028)	-0.165	(0.325)	0.025	(0.141)
8	1939	0.202***	(0.057)	0.919	(0.808)	0.092	(0.139)
7	1940	0.141***	(0.026)	1.128***	(0.299)	0.194**	(0.069)
6	1941	0.182**	(0.077)	0.455	(0.510)	0.078	(0.092)
5	1942	0.128***	(0.042)	0.935**	(0.380)	0.055	(0.090)
4	1943	0.161***	(0.043)	0.812	(0.576)	0.050	(0.115)
3	1944	0.164***	(0.048)	1.154***	(0.278)	0.198*	(0.097)
2	1945	0.211**	(0.078)	1.331	(0.831)	0.186	(0.121)

Notes: Coefficients π_τ from Equation (2), one regression per outcome, same sample as Figure 2. Standard errors clustered by county. The 1931 cohort (age 16 in 1947) is the reference and is set to zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B Reproducibility

- The IPUMS extract is not redistributable; see README.
- One command regenerates every number and figure: `cd` to the repo root, `do run.do`
- `estout` is vendored in `code/ado/estout` so no package installation is needed
- folder map below

```
run.do                    master; sets paths, runs code/01-06 in order
code/01_build_data.do     IPUMS extract  -> data/clean/antman_cortes_clean.dta
code/02_figure1.do        cohort means   -> output/figures/fig1_*
code/03_table1.do         Table 1        -> output/tables/table1_summary.*
code/04_table2.do         Table 2 A/B/C  -> output/tables/table2_panel*.*
code/05_event_study.do    Figures 2/A1/A2, output/tables/eventstudy_coefs.*
code/06_prose_numbers.do  numbers cited in the text -> output/tables/prose_numbers.*
code/lib/yrsed.do         EDUCD -> years of schooling crosswalk
```

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